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Försättsblad Prov Original

Kurskod	Provkod	Tentamensdatum
DT081A	T101	2025-10-28
Kursnamn	Datateknik AV, Signal- och bildbehandling	
Provnamn	Tentamen	
Ort	Sundsvall	
Termin		
Ämne		

6895642268

Utskriven 2025-12-30 kl. 08:49:24

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Department of
Computer and Electrical Engineering (DET)
Roger Olsson (Roger.Olsson@miun.se), 010 142 8698
Mårten Sjöström (Marten.Sjostrom@miun.se), 010 142 8836

2025-10-28. Examination in

DT081A – Signal and Image Processing, 6 hp

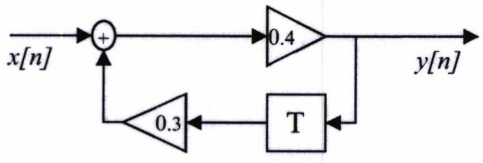
Allowed aids	Dictionary, non-programmable calculator and drawing tools (e.g ruler), Physics Handbook, TeFyMa, and formula collection booklet for mathematics/physics courses at MIUN. A collection of formulas is also presented at the end of this exam.
Miscellaneous	Don't forget to: 1. Submit one, or several pages, per solved assignment and NOT several solved assignments per page. 2. Order the handed in pages with the solved assignments in ascending order. 3. Describe and clearly motivate all solutions, assumptions, programming constructs, etc.. 4. Attach all neatly drawn figures that you might want to refer to in a solution. **Failing to follow the above instructions will render point reduction.**

Good Luck!

1. What do the following signal processing concepts mean? Be clear, concise, and relevant.

a) Sampling theorem (1p)
b) Aliasing (1p)
c) Infinite Impulse Response (IIR) (1p)
d) Spectrum (1p)
e) Digital signal (1p)
f) Separability (1p)

2. A time discrete system is defined as below.



a) Using the block diagram, explain whether the system is FIR or IIR, causal or non-causal, and justify your answer based on its structure. (2p)
b) Derive the system's difference equation from the block diagram. (2p)

3. Given a discrete-time system described by the difference equation

$$y(n) + y(n-1) = x(n) + 0.5y(n-2) - 0.3\mathrm{x}(\mathrm{n}-1)$$

a) Identify the type of response that the system produces by the input

$$x(n) = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}$$ (1p)

b) Calculate the first 3 samples of the output signal $y(0), y(1), y(2)$ by the input

$$x[n] = \begin{cases} 0.6^n \cdot \cos[n], & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$ (2p)

c) Another system has the frequency function

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} = \frac{1}{1-e^{-j\Omega}-0.5e^{-2j\Omega}} = \frac{e^{2j\Omega}}{e^{2j\Omega}-e^{j\Omega}-0.5}.$$

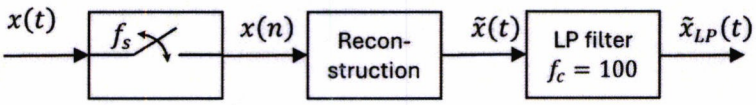
Sketch the amplitude magnification $\left|H(e^{j\Omega})\right|$ for frequency $0 \leq \Omega \leq 2\pi$. (3p)

4. The signal $x(t) = 2\cos(60\pi t + 0.2) + 3\sin(30\pi t + 0.1)$ is sampled and reconstructed according to the figure below.

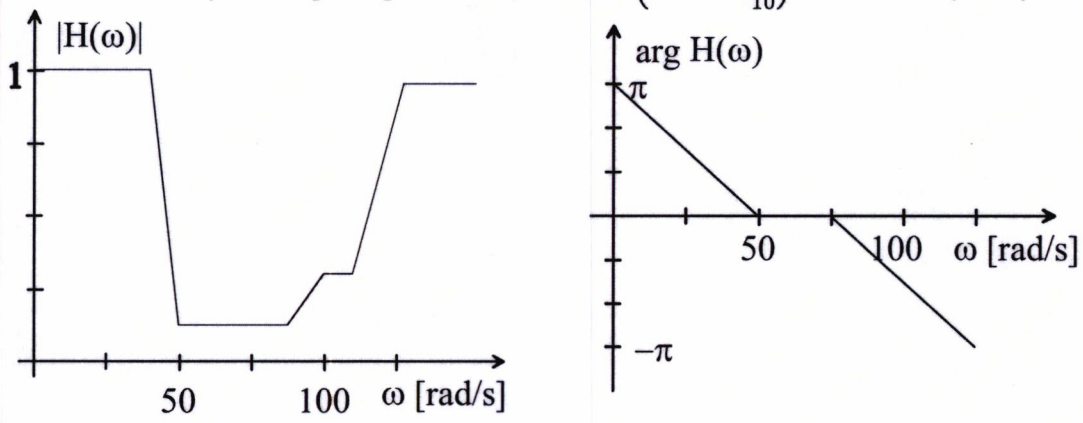
a) What is the lowest sampling frequency f_s [Hz] to enable a full reconstruction of the signal with the same frequency contents? (1p)

b) If the signal is sampled with $f_s = 100\ Hz$, reconstructed and filtered with an ideal LP-filter with cut-off frequency $f_c = 100\ Hz$, what frequencies does the output signal $\tilde{x}_{LP}(t)$ then contain. [Hint: there is aliasing] (2p)

c) Draw the amplitude spectrum for the signal $x(t)$ and sketch (in the same diagram) the frequency response for a real (non-ideal) antialiasing filter that allow full reconstruction of the signal. (1p)

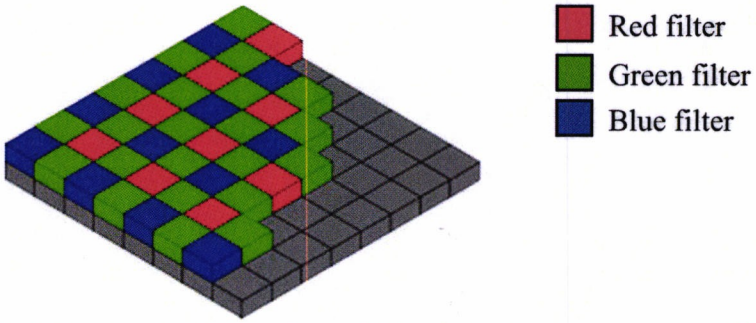


5. A linear, time-invariant (LTI) system is defined by the amplitude and phase spectra in the figure below. The system input signal is $x(t) = 3\cos\left(10\pi t + \frac{\pi}{10}\right) + 1 + 4\sin(23\pi t)$.



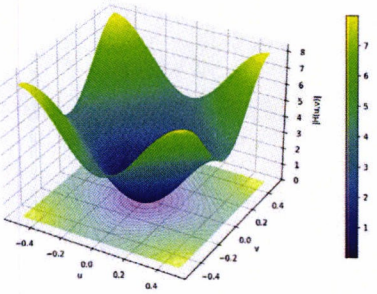
a) Decide the output signal $y(t)$. (2p)
b) What kind of filter is the system? (1p)

6. Digital camera sensors typically capture only one color component (red, green, or blue) per pixel using a Bayer filter array like the one shown below. To reconstruct a full-color image, the missing color values at each pixel are interpolated from neighboring pixels.



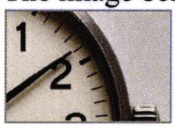
a) Explain why the demosaicing (debayering) process necessarily involves a trade-off between color fidelity and spatial resolution. (2p)
b) Describe in general terms how a simple demosaicing algorithm reconstructs full RGB values from a Bayer-pattern image. *Hint: refer to how missing color values at each pixel are handled.* (2p)

7. The figure below shows the magnitude of the frequency response, $|H(u,v)|$, of a two-dimensional linear system.

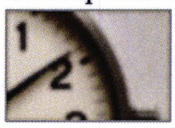


a) From the shape of $|H(u,v)|$, describe how the filter treats different spatial frequencies. (2p)

b) The image below is the input to the filter.



Which one of the following four candidate output images (i–iv) most likely corresponds to the filtered result? Motivate your choice



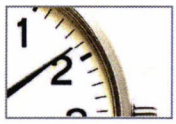
i)



ii)



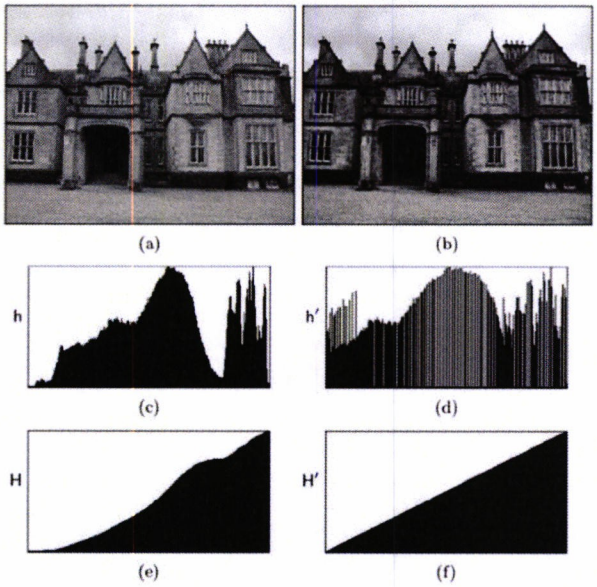
iii)



iv)

(3p)

8. Contrast enhancement aims to improve the visual appearance of an image by modifying its intensity distribution. The figure below shows an original image (a) and an enhanced version (b). Plots (c–f) show different representations related to these two images. Describe what information each of the plots (c–f) conveys and explain, using equations where relevant, how they are involved in producing the enhanced image (b). (5p)



Collection of Formulas

Base Signals

$$x(t), (continous)$$

$$x[n], (discrete)$$

Stochastic

$$w(n) \sim \mathcal{N}(0, \sigma^2)$$

Linearity

$$ax_1 + bx_2 \rightarrow ay_1 + by_2$$

Time-Shift

$$x(t - t_0) \rightarrow y(t - t_0)$$

Causality

$$h(t) = 0,\ t < 0$$

Nyquist-Shannon Sampling Theorem

$$f_s \geq 2f_{max}$$

Convolution

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau \text{ (continuous)}$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k] \text{ (discrete)}$$

Euler's Formula

$$e^{jx} = \cos(x) + j\sin(x)$$

Fourier Series

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}$$

Fourier Transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

DTFT

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

DFT

$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

Time/Space Shift Property

$$x(t - t_0) \leftrightarrow X(\omega) e^{-j\omega t_0} \text{ (continuous)}$$

$$x[n - n_0] \leftrightarrow X(e^{j\Omega}) e^{-j\Omega n_0} \text{ (discrete)}$$

Spectrogram

$$S(t, \omega) = |X(t, \omega)|^2$$

FIR

$$y[n] = \sum_{k=0}^{M} b_k x[n - k]$$

IIR

$$y[n] = \sum_{k=1}^{N} a_k y[n - k] + \sum_{k=0}^{M} b_k x[n - k]$$

Frequency Response

$$H(e^{j\Omega}) = |H(e^{j\Omega})| e^{j \arg H(e^{j\Omega})}$$

PSF Basic

$$h(x, \alpha, y, \beta)$$

Shift-Invariant

$$g(\alpha, \beta) = \sum_{x} \sum_{y} f(x, y) h(\alpha - x, \beta - y)$$

DCT Basis

$$U_{x\alpha} = \sqrt{\frac{2}{N}} \cos\left(\frac{\pi(2x + 1)\alpha}{2N}\right)$$

RGB to YCbCr

$$\begin{bmatrix} Y' \\ Cb \\ Cr \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.500 \\ 0.500 & -0.419 & -0.081 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Gamma Correction

$$C' = C^{1/\gamma}$$